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DBMS TASK - 9 REPORT

Title: CRUD operations in Graph databases

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TASK 9

CRUD operations in Graph databases

AIM:

To perform CRUD operations like creating, inserting, querying, finding, deleting operations on graph spaces.

The steps to get started with Neo4j's Aura Graph Database:

Step1: Copy and paste the following link into your web browser:

<https://neo4j.com/cloud/platform/aura-graph-database/?ref=docs-get-started-dropdown>

Step2: Click on "Start Free."

Step3: Choose the option to "Continue with Google."

Step4: Click the "Open" button.

Step5: After clicking "Open," a text file will be automatically downloaded. This file contains your user ID and password details.

Step6: Copy the password from the downloaded text file and paste it where required.

Step7: Close the "Get started with Neo4j with beginner guides" if it's open.

Step8: You're now ready to begin practicing with the Graph Database.

Create Node with Properties

Properties are the key-value pairs using which a node stores data. Create a node with properties using the CREATE clause and need to specify these properties separated by commas within the flower braces "{ }".

Syntax

```
CREATE (node:label{ key1: value, key2: value, . . . . . }) return node
```

To verify the creation of the node, type and execute the following query in the dollar prompt.

Syntax:

```
MATCH (n) RETURN n
```

Creating Relationships

To create a relationship using the CREATE clause and specify relationship within the square braces “[]” depending on the direction of the relationship it is placed between hyphen “ - ” and arrow “ → ” as shown in the following syntax.

Syntax:

```
CREATE (node1)-[:RelationshipType]->(node2)
```

Syntax:

```
MATCH (a:LabeofNode1), (b:LabeofNode2)
```

```
WHERE a.name = "nameofnode1" AND b.name = " nameofnode2"
```

```
CREATE (a)-[: Relation]->(b) RETURN a,b
```

Deleting a Particular Node

To delete a particular node and need to specify the details of the node in the place of “n” in the above query.

Syntax:

```
MATCH (node:label {properties . . . . . }) DELETE node
```

Create a graph database for student course registration, create student and dept node and insert values of properties.

Create a CrickerBoard Node:

```
create(cb:CricketBoard{BoardID:'BID01',Name:'Chennai Cricket Board', Address:'Chennai',  
Phone:9988776699}) returncb
```

Create Team Nodes:

```
create(t1:Team{teamID:'CCB01',BoardID:'BID01',name:'ABS EXPRESS',  
Coach:'G.D.RAMESH', Captain:'SAMPATH KUMAR'}) return t1
```

```
create(t2:Team{teamID:'CCB02',BoardID:'BID01',name:'AVG EXPRESS',Coach:  
'T.KARTHIKH', Captain:'Y.JOHN'}) return t2
```

Create Player Nodes:

```
create(p1:Player{PlayerID:'1',TeamID:'CCB01',Name:'Raj',Age:23,DateofBirth:'29-JUN-1996',  
PlayingRole:'Bowler',email:'rajn@gmail.com'}) return p1
```

```
create(p2:Player{PlayerID:'33',TeamID:'CCB01',Name:'Anand',Age:23,DateofBirth:'02-JAN-19  
99', PlayingRole:'Batsman',email:'balajid@gmail.comm'}) return p2
```

```
create(p3:Player{PlayerID:'65',TeamID:'CCB02',Name:'Suresh',Age:27,DateofBirth:'02-JUN-1996', PlayingRole:'Batsman',email:'sureshd@gmail.comm'}) return p3
```

```
create(p4:Player{PlayerID:'75',TeamID:'CCB02',Name:'Rohit',Age:33,DateofBirth:'02-JUN-1991', PlayingRole:'Batsman',email:'srohit@gmail.comm'}) return p4
```

Creating Relationship among CricketBoard and Teams:

```
match(cb:CricketBoard{BoardID:'BID01'}),(t1:Team{teamID:'CCB01'}) create(cb)-[r:has]->(t1) return cb,r,t1
```

```
match(cb:CricketBoard{BoardID:'BID01'}),(t2:Team{teamID:'CCB02'}) create(cb)-[r:has]->(t2) return cb,r,t2
```

Creating Relationship among Players and Teams:

```
match(p1:Player{PlayerID:'1'}),(t1:Team{teamID:'CCB01'}) create(p1)-[r1:playfor]->(t1) return p1,r1,t1
```

```
match(p2:Player{PlayerID:'33'}),(t1:Team{teamID:'CCB01'}) create(p2)-[r2:playfor]->(t1) return p2,r2,t1
```

```
match(p3:Player{PlayerID:'65'}),(t2:Team{teamID:'CCB02'}) create(p3)-[r3:playfor]->(t2) return p3,r3,t2
```

```
match(p4:Player{PlayerID:'75'}),(t2:Team{teamID:'CCB02'}) create(p3)-[r4:playfor]->(t2) return p4,r4,t2
```

Display All nodes: match(n) return n

Output:

The screenshot displays the Neo4j Aura workspace interface. On the left, the 'Database Information' sidebar shows 8 nodes (CricketBoard, Player, Team) and 7 relationships (has, playfor). The central query editor contains the Cypher query: `neo4j $ match(n) return n`. The 'Results Overview' panel on the right shows the query results: 8 nodes (1 CricketBoard, 4 Players, 2 Teams). The graph visualization in the center shows the nodes and their relationships. The Windows taskbar is visible at the bottom.

OUTPUT:

The screenshot shows the Neo4j Aura workspace interface. The left sidebar displays 'Database Information' with 8 nodes (CricketBoard, Player, Team) and 7 relationships (has, playfor). The main query editor shows the query `neo4j$ match(n) return n`. The graph view displays a network of nodes and relationships. The 'Node Details' panel on the right shows the details for a 'Team' node with the following properties:

Property	Value
<id>	6
BoardID	"BID01"
teamID	"CCB01"
name	"ABS EXPRESS"
Captain	"SAMPATH KUMAR"
Coach	"G.D.RAMESH"

Retrieve particular player details:

`match(p:Player{PlayerID:'33'}) return p`

The screenshot shows the Neo4j Aura workspace interface. The left sidebar displays 'Database Information' with 8 nodes (CricketBoard, Player, Team) and 7 relationships (has, playfor). The main query editor shows the query `neo4j$ match(p:Player{PlayerID:'33'}) return p`. The graph view displays a single node labeled 'Anand'. The 'Node Details' panel on the right shows the details for a 'Player' node with the following properties:

Property	Value
<id>	1
PlayerID	"33"
PlayingRole	"Batsman"
DateofBirth	"02-JAN-1999"
TeamID	"CCB01"
email	"balajid@gmail.com"
Age	23
Name	"Anand"

Update particular player details:

`match(p:Player{PlayerID:'1'}) set p.age=27 return p`

Output:

The screenshot displays the Neo4j Aura workspace interface. On the left, the 'Database Information' panel shows 8 nodes (CricketBoard, Player, Team) and 7 relationships (has, playfor). The central query editor contains the Cypher query: `neo4j $ match(p:Player{PlayerID:'1'}) set p.age=27 return p`. Below the query, a graph visualization shows a single node labeled 'Raj'. On the right, the 'Node Details' panel for the 'Player' node shows the following properties: <id> 8, PlayerID '1', PlayingRole 'Bowler', DateofBirth '29-JUN-1996', TeamID 'CCB01', age 27, email 'rajn@gmail.com', Age 23, and Name 'Raj'. The bottom status bar indicates the last update was at 10:10:55 pm.

Delete particular player from the team:

`match(p:Player{PlayerID:'33'}) delete p`

The screenshot displays the Neo4j Aura workspace interface. The central query editor contains the Cypher query: `neo4j $ match(p:Player{PlayerID:'33'}) delete p`. Below the query, an error message is shown: **Neo.ClientError.Schema.ConstraintValidationFailed** with the text: 'Cannot delete node<1>, because it still has relationships. To delete this node, you must first delete its relationships.' The left sidebar and top navigation bar are consistent with the previous screenshot.

Result:

Thus the CRUD operations like creating, inserting, querying, finding, deleting operations on graph spaces were executed successfully.